

The Bridge-Contract Criterion: A Form Audit for Transport Arguments, with an Instantiation in the abc Literature

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Abstract

Some proof claims resist line-by-line review not because individual steps are opaque, but because their load-bearing core is a *transport*: a quantity migrates between theoretical frameworks, and the claim’s numerical content rests on the admissibility of that migration. A related class rests on *global-only* statements: assertions available only after summation over all places, with no place-wise access. We propose a seven-component bridge contract and a versioned ledger audit. We report a retrospective two-case feasibility illustration of the frozen v1-T transport ledger and one substantive selected-rows instantiation of the criterion in the abc literature (the v1-F rows exercised there are named; the full-matrix coding of all rows is deferred). The paper contains no RH-side verdict; RH uniformity and global-only positivity claims are retained as an explicitly untested outlook. The criterion judges contract *form*, not mathematical truth: a served ledger is necessary, not sufficient, and the auditor tests the presence and addressability of an obligation, never its mathematical validity. Its value is that failures localize — the audit names the missing clause at a stated row instead of a diffuse doubt. The current research state aligns the conditional C7 illustration with all five hypotheses of the unchanged contract definition, after a counterexample to the earlier incomplete diagram. Later developments in Parts 4–6 are distinguished from the historical audit pins: same-field rigidity permits global scaling, while coordinate, field, period and volume bridges remain open. The two-case raw matrix remains a version-scoped illustration, not a reliability estimate.

Contents

1	Introduction	4
2	The Criterion	5
2.1	The Bridge Contract	5
2.2	Reading the Domain Expressions	8
2.3	Ledger Form and Verdict Grammar	9
2.4	Crosswalk Between the Two Instrument Versions	11
2.5	Scope of the Criterion	11
3	Related Auditing Structures	12
3.1	Proof Auditing and Peer-Review Methodology	12
3.2	Formal Verification and Lean	12
3.3	Certificate Interfaces in Computer Science	13
3.4	Assurance Cases and Goal Structuring Notation	13
3.5	Error Analyses and Controversy Methodology	14
3.6	Position of the Criterion	14
4	Two Complementary Gate Types	15
5	A Protocol-Frozen Retrospective Two-Case Illustration (v1-T)	15
5.1	Design and Instrument Version	18
5.2	The Retrospective Known-Gap Case: The 1993 Fermat Announcement	19
5.3	The Accepted Control: An Accepted Transport Step	19
5.4	The Complete Code Matrix	19
5.5	Anchoring and Availability	20
5.6	What the Illustration Shows—And What It Does Not	20
6	The abc Instantiation	21
6.1	The Gate Case	21
6.2	The Audited Substep of the Original Argument	21
6.3	A Comparative Discrimination Case: Joshi’s ATS Program	24
6.4	Inheritance: What Travels with a Citation	25
6.5	An External Point of Comparison	27
7	Outlook: RH Uniformity Claims	28
8	A Related Structure: Global-Only Claims	28
8.1	Current Developments in Parts 4–6	29
9	Limitations and Fairness	30

1 Introduction

Two classes of proof claims resist classical line-by-line review in a characteristic way. The first are *transport claims*: a quantity is constructed in one framework and its numerical reading is asserted to land in a target region fixed in another, with the claim’s content resting on the admissibility of the migration rather than on any single computational step. The documented precedent is the disputed transport step of Mochizuki’s Corollary 3.12, whose ten-year controversy consists, on our reading, not of disagreement about individual lines but of disagreement about what the transport *owes* [11]. The second are *global-only claims*: assertions available only after summation over all places, with no place-wise access — a shape that occurs both in the abc literature and in positivity approaches to the Riemann Hypothesis.

This paper proposes a checkable form criterion for both classes: the *bridge contract*, a list of seven contract components that a transport must serve before its numerical conclusion may be drawn, together with a ledger-style audit protocol with a fixed verdict grammar. The criterion grew out of the audits of this series — the gap localization of Part 1 [11] and the certificate instrument of Part 3 [12] — and has since been applied, in the later parts, to the principal published repair program [13, 14, 15]. Here we present the criterion itself as a transferable method: its definition (Section 2), its position among existing auditing structures (Section 3), a protocol-frozen retrospective two-case illustration of the frozen transport ledger (Section 5), and one selected-rows instantiation of the criterion on the abc transport step (Section 6). The companion landscape paper of the program records how the criterion sits among the routes it audits [20].

Two limits of scope are stated here rather than left to be inferred. First, the instrument is *versioned*: the full ledger of Section 2.3 and the six-row transport ledger used for the retrospective illustration are two distinct instruments with a published crosswalk, and identical numeric row suffixes do not imply identical semantics. Second, this paper reports exactly one instantiation, in the abc literature, and that instantiation is *selected-rows*: the rows exercised are named in Section 6, and a coding of the complete v1-F matrix for these cases is deferred to a later version. By an *instantiation* we mean throughout the application of the criterion to one dispute complex — here the abc transport dispute, comprising the original argument and its repair program as a single complex with two audited profiles. Uniformity claims toward the Riemann Hypothesis and global-only positivity claims appear as an explicitly untested outlook (Sections 7 and 8); no RH-side verdict is drawn anywhere below.

To keep the counting unambiguous, the cases treated in this paper are named by kind and not by ordinal throughout: a *retrospective known-gap case* and an *accepted control* (Section 5), the *abc instantiation* with a *comparative discrimination case* and a *downstream inheritance example*

(Section 6), an *external comparison* (Section 6.5), and the *RH / global-only outlook* (Sections 7 and 8).

Throughout, one boundary is maintained without exception: the criterion judges *contract form*, not mathematical truth. A fully served ledger is necessary for the admissibility of a transport reading, never sufficient for the underlying claim; an unserved row localizes an obligation, it does not refute a theorem. The auditor tests whether an obligation is present and addressable in a named version of a text, never whether the mathematics it names is correct. The value we claim for the method is exactly this localization: audits that end in a named missing clause rather than a diffuse doubt.

2 The Criterion

2.1 The Bridge Contract

The situation we axiomatize is the following. A proof asserts a numerical conclusion of the form “the reading of an object constructed in framework A lands in a target region fixed in framework B ”, where the two frameworks do not share a common evaluation by construction. The assertion is mediated by a *transport*: labels, containers, or functors that relate the two sides. The criterion asks what the transport must *contractually* provide before the numerical conclusion may be drawn.

Definition 2.1 (Bridge contract). *Let a claim relate a source object to a numerical target region. A bridge contract Br for this claim consists of the following seven components.*

1. **Source object with readout** (X_q, ν_q, i_q) : *the source-side domain X_q , a concrete numerical readout functional $\nu_q: X_q \rightarrow V$ into the ordered value space of Component 2, and the typed map $i_q: X_q \rightarrow C$ into the container of Component 3 (in the motivating case: a log-volumetric reading of the q -pilot).*
2. **Target object with target readout and fixed target region** $(X_\Theta, \nu_\Theta, (V, \preceq), H_\Theta)$: *the target domain X_Θ , a target readout $\nu_\Theta: X_\Theta \rightarrow V$, an ordered value space $(V, \preceq)$ shared by both readouts, and a target region $H_\Theta \subseteq V$ fixed in advance — a lower set $H_\Theta = \{v \in V : v \preceq \tau_\Theta\}$ or the corresponding upper set, with the threshold $\tau_\Theta \in V$ named before the source value is inspected. The typed map $i_\Theta: X_\Theta \rightarrow C$ belongs to Component 3. In the motivating case $V = \mathbb{R}$ with its usual order and $H_\Theta = \mathbb{R}_{\leq -|\log \Theta|}$.*
3. **Typed common container** $(C, \preceq_C, i_q: X_q \rightarrow C, i_\Theta: X_\Theta \rightarrow C)$: *a common receptacle equipped with the order relation $\preceq_C$ used by the claim (a partial order, or the preorder the source itself specifies, stated*

with its carrier), together with explicitly typed maps from both sides, each given with its domain and codomain. A container in which both objects merely cohabit does not by itself carry scale, and a container without a stated order carries no order claim.

4. **Admissible linkage** (L): label or linkage data that specify an allowed identification, not mere homonymy. Sharing a name across frameworks is not a mathematical relation.
5. **Separated indeterminacy budget** (I): an explicit budget distinguishing absorbable renormalization, blurring, and non-absorbed residual load. An indeterminacy that is invoked to absorb an obstruction must be booked against this budget, not against the container.
6. **Compatible common evaluation** (ν_C). A typed map $\nu_C: C \rightarrow (V, \preceq)$ is supplied together with the compatibility identities required by the claim, in particular $\nu_C \circ i_q = \nu_q$ on the stated source domain and, where the target is represented by a target readout ν_Θ , $\nu_C \circ i_\Theta = \nu_\Theta$ on the stated target domain. These compatibility clauses are mandatory; the mere existence of a numerical map on C never discharges this component.
7. **Addressed order lemma.** A named lemma states the hypotheses under which the admissible linkage yields an order relation in C , states the monotonicity and target-region hypotheses, and derives the required target membership. For a lower target set $H_\Theta = \{v \in V : v \preceq \tau_\Theta\}$, a typical form is: if (H1) $L(x, y)$ is admissible, (H2) $i_q(x) \preceq_C i_\Theta(y)$, (H3) ν_C is monotone on such linked pairs, (H4) the compatibility clauses of Component 6 hold, and (H5) $\nu_C(i_\Theta(y)) \preceq \tau_\Theta$, then $\nu_q(x) \in H_\Theta$. Hypothesis (H2) is never a free assumption: the contract requires the text either to derive it from the linkage data of Component 4 in an addressed sublemma, or to prove it separately — in both cases with its own proof address. The same anchoring requirement applies to every load-bearing hypothesis of the lemma.

A contract audit records this component as textually served only when the lemma, its hypotheses, its claimed applicability at the audited step, and a proof address are present and inspectable in the named version. The auditor tests presence and addressability, not mathematical validity. A served row therefore does not certify that the lemma or its proof is correct.

Components 6 and 7 are the load-bearing clauses: without them, common-container cohabitation together with shared labels yields only qualitative nearness. In the precedent cases audited below they are also where the heaviest findings fall — but not the only rows that carry findings, and the ledger reports the full profile rather than a single surviving row.

Figure 1 illustrates the conceptual architecture of a bridge contract, mapping the interaction of the seven canonical components between the disjoint

source and target frameworks. Table 1 summarizes the seven canonical components, their mathematical signatures, the required verification obligations, and the characteristic failure modes.

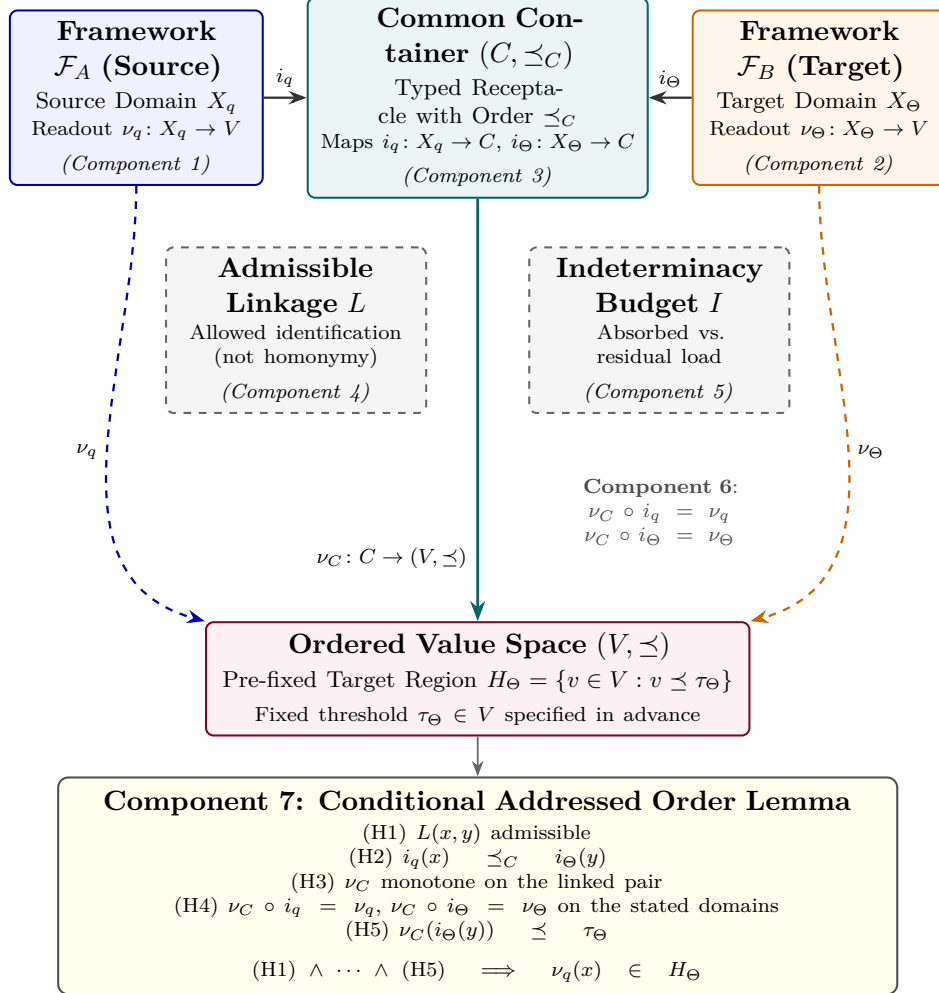


Figure 1: Conditional bridge-contract illustration for the lower target set. All five hypotheses are required; (H2) and every load-bearing obligation need their own proof addresses. The diagram illustrates the contract and does not certify the correctness of a source proof.

Development of the C7 illustration. The earlier diagram omitted the target bound (H5) and replaced compatibility (H4) by a bounded residual load. Its implication fails in the identity model: take every domain and carrier to be $\mathbb{R}$, every displayed map to be the identity, the usual order, $x = y = 1$, admissible $L(x, y)$, residual load 0, $\tau_\Theta = 0$, and $H_\Theta = (-\infty, 0]$. The earlier

displayed premises hold, including compatibility, but $\nu_q(x) = 1 \notin H_\Theta$. This refutes the incomplete diagram implication, not Definition 2.1 with (H1)–(H5). Figure 1 now displays those five conditions separately; the illustration does not certify a source proof or establish experimental validation of the method.

Table 1: The seven canonical components of a bridge contract Br and their verification obligations.

Component	Mathematical Signature	Verification Obligation	Failure Mode / Gap Typology
C1: Source readout	(X_q, ν_q, i_q)	Explicit domain, codomain, and evaluation $\nu_q : X_q \rightarrow V$	Untyped source, implicit readout
C2: Target region	$(X_\Theta, \nu_\Theta, (V, \preceq), H_\Theta)$	Value space $(V, \preceq)$ and target region H_Θ fixed <i>ex ante</i>	Moving threshold, <i>post hoc</i> region shift
C3: Typed container	$(C, \preceq_C, i_q, i_\Theta)$	Ambient carrier C with partial order/preorder $\preceq_C$ and typed maps	Scale-free cohabitation, unordered carrier
C4: Admissible linkage	$L(x, y)$	Allowed identification rule specified by theorem/hypothesis	Nominal identity, ungrounded homonymy
C5: Indeterminacy budget	$(I, \text{Renorm}, \text{Blur}, \text{Res})$	Explicit separation of quotiented group actions vs. residual load	Residual charged to container, unbooked drift
C6: Compatible evaluation	$\nu_C : C \rightarrow V$	Commutativity identities $\nu_C \circ i_q = \nu_q$ and $\nu_C \circ i_\Theta = \nu_\Theta$	Coincidence assumed, missing commutation
C7: Addressed order lemma	Lemma Ord(...), (H1)–(H5)	Monotonicity of ν_C , derivation of (H2) $i_q(x) \preceq_C i_\Theta(y)$	“Follows formally”, unanchored order jump

2.2 Reading the Domain Expressions

Four expressions in Definition 2.1 are meant generically, and are fixed here so that the contract can be instantiated outside the motivating case. An identification is *allowed* when it is admitted by a named hypothesis or a named class of theorems — shared notation is not a license. An indeterminacy is *absorbable* when it can be quotiented out by an explicitly stated equivalence or invariance relation. The *residual load* is the term that remains after every permitted quotient has been taken; it is booked against the budget of Component 5 and never against the container. A *second mode* is an alternative branch, normalization, or sign convention that can change the target value or

the order relation, and whose presence triggers the corresponding ledger row. The inter-universal Teichmüller vocabulary of the motivating case supplies examples of these expressions, not their definitions.

2.3 Ledger Form and Verdict Grammar

Instrument versions. We distinguish two version-scoped instruments. **BCM-Ledger v1-F** is the full ledger used to operationalize the general seven-component criterion in this paper. **BCM-Ledger v1-T** is the six-row transport instrument frozen on 15 August 2026 before the two historical source-coding passes. The retrospective two-case illustration in Section 5 concerns v1-T only. Row labels are scoped by the version prefix; identical numeric suffixes do not imply identical semantics. The alignment table below is a crosswalk, not an assertion that the two instruments are interchangeable.

The contract is audited by a fixed ledger of test rows. Each row receives one of three verdicts — *delivers* / *does not deliver* / *partially delivers* — together with a confidence qualifier and a source anchor (page/section citation of the audited text). Several rows are split, a lesson learned from the audits themselves: without the split, two distinct findings collapse into one word. Table 2 is normative for v1-F. “Primary component” names the localized contract obligation; dependencies are listed transparently.

Each active *core* row (F-R0 through F-R7) has exactly one primary contract component. Supporting dependencies may be listed, but a negative verdict is reported at the primary component named in the crosswalk. The extension row F-R8 is the sole exception: by its E8 definition it maps to no component, and a triggered F-R8 finding is reported at the extension label E8 itself, never at a contract component. F-R8 is not a universal core row; it is activated only by the stated multi-mode trigger and is otherwise coded X.

Coding uses five raw codes: explicitly discharged (E+), explicitly not dischargeable (E−), dischargeable only via a documented reconstruction (I), not addressed (N), and out of scope (X), the last admissible only with an explicit scope or trigger justification. I and N are form findings, not error findings. Raw codes are never erased by the public verdict grammar: E+ maps to delivers; I maps to partially delivers by documented reconstruction; E− and N map to does not deliver, retaining the distinction between explicit openness and non-address; X is out of scope. Inter-coder pairs remain reported as pairs. A row coded X carries no verdict and is not counted in the denominator. Where a row is coded twice, the double-coding rule used in the retrospective illustration is stated in Section 5; it is a rule of that instrument version and not part of v1-F.

Table 2: The full BCM-Ledger v1-F: core audit rows, primary contract components, and observable test criteria.

v1-F row	Primary	Observable test of service
F-R0	C1 (dep. C2)	Source, target, readouts, value space and target region are separately typed with domain and codomain.
F-R1	C3	A common container and both typed embeddings or maps are named.
F-R2a	C4 (dep. C6)	For each intra-link step the transfer law is given with an address and a visible readout effect; “no operation” yields $\mathbf{X}$, not automatically $\mathbf{E+}$.
F-R2b	C6 (dep. C3)	The cross-container step carries a visible conversion factor or compatibility equation; mere convention or coincidence is at most $\mathbf{I}$.
F-R3	C2	H_Θ resp. τ_Θ is fixed before the source value is inspected; no <i>post hoc</i> moving.
F-R4a	C5	Every non-neutral load or compensation is quantified and assigned to a budget category.
F-R4b	C5	The load is booked at the bridge step itself, with carrier, sign and residual term; downstream of a presupposed conclusion does not suffice.
F-R5	C5	Absorbable renormalization, blurring and non-absorbed residual are kept apart; the residual is not charged to the container.
F-R6a	C6	The common evaluation and all compatibility identities the claim needs are present in the text and addressed.
F-R6b	C7	The named order lemma lists its hypotheses, asserts their applicability at the step, and carries a concrete proof address; in addition, each load-bearing hypothesis — in particular (H2) — carries its own anchor (derivation from the linkage data, or separate proof). The auditor tests presence and addressability of these anchors, never their validity.
F-R7	C7 (dep. C2)	The statement actually derived is membership in H_Θ ; approximation, proximity or asymptotic approach is not relabelled as membership.
F-R8	<i>extension row E8</i>	E8 (definition). E8 is a domain extension outside the seven-component crosswalk: it maps to no contract component, and it is part of v1-F only as a triggered diagnostic. <i>Trigger:</i> several modes, branches, normalizations, or a sign- or order-relevant alternative mode are present in the claim’s domain model. Then all modes must be visible and either covered by the order lemma or excluded with reasons (this is where E8 touches C5 and C7 without belonging to them). Without the trigger the row is necessarily $\mathbf{X}$.

2.4 Crosswalk Between the Two Instrument Versions

The frozen transport ledger v1-T is functionally close to parts of v1-F but not identical to them. The alignment is published so that the retrospective illustration of Section 5 can be read against the full instrument without either being substituted for the other.

Table 3: Crosswalk between the frozen transport instrument (BCM-Ledger v1-T) and the full instrument (BCM-Ledger v1-F).

v1-T row	Frozen semantics	Nearest in v1-F
T-R2a	transfer law explicit	F-R2a / C4
T-R2b	visible conversion factor	F-R2b and F-R6a / C6
T-R4a	compensation mechanism named and located	F-R4a / C5
T-R4b	existence and well-definedness proven	<i>no direct equivalent</i> [†]
T-R6	second modes and cases visible and symmetric	F-R8 / triggered E8
T-R8	chain-closing compatibility derived, locally or globally	F-R6a, F-R6b / C6–C7

[†] T-R4b has no direct v1-F equivalent: the nearest neighbor is F-R4b (booking of the load at the bridge step), but the frozen existence/well-definedness semantics of T-R4b is not itself carried by any v1-F core row. The crosswalk records proximity, not identity.

Table 3 shows functional proximity, not identity. Note in particular that the numeric suffixes cross: v1-T’s second-mode row is T-R6, whereas the second-mode row of v1-F is F-R8. This is the reason the version prefix is carried on every row label in this paper.

2.5 Scope of the Criterion

The criterion judges *contract form*, not truth. What a fully served ledger establishes is textual contract completeness for the transport reading of a selected step; it is not a proof of the underlying claim, and it does not certify the mathematics that the served rows name. Conversely, an unserved row does not refute the claim — it localizes the burden: the row names the clause a complete argument must serve. This asymmetry is the criterion’s purpose. The unit of audit is correspondingly narrow: the ledger decomposes one selected transport dispute into named obligations; it does not reduce the historical controversy as a whole. What an audit yields is a profile of named obligations at a selected disputed transport step, with weights — which is where mathematical work, rather than exegesis, can then be invested.

3 Related Auditing Structures

Five bodies of work bear on the criterion of Section 2: normative accounts of when an omission in a mathematical proof is acceptable; machine formalization of deep mathematics; certificate interfaces in computer science; structured assurance argumentation in safety engineering; and the methodology of documented mathematical disagreement. Two observations organize what follows. First, the closest heuristic analogue among the structures compared here is the certifying-algorithms tradition [28], from which the criterion differs in the direction of the certificate and in what the check guarantees (Section 3.3). Second, within the corpus surveyed here we are aware of no independent, peer-reviewed case analysis of the abc/IUT controversy — a gap that persists and that the present work does not close, its contribution lying on the level of contract form (Sections 3.5 and 2.5).

3.1 Proof Auditing and Peer-Review Methodology

Andersen [3] asks when a gap in a published proof may be treated as acceptable and gives normative criteria for the substance of what is omitted. The bridge contract answers a different question: whether the contract components of Section 2.1 are served in the presentation at all. The two assessments are independent — a fully served ledger is a statement about form, not about the truth of the transported claim. De Millo–Lipton–Perlis [10] describe proof acceptance as a social process carried by communal reading. The ledger is independent of community uptake and of a particular institutional peer-review procedure; it is not auditor-independent, as shown by the reported coding dissent (Section 5). Andersen [2] documents what referees in mathematics actually check — a descriptive account of an existing process; the criterion is an instrument that can be applied outside any such process. Avigad [5] treats the reliability of informal mathematical inference in general; the criterion addresses one special case within that range: a transport step whose conclusion is a numerical membership claim.

3.2 Formal Verification and Lean

Scholze’s Liquid Tensor Experiment [36] deliberately submitted a contested result of his own to a machine-checkable contract. It answers the truth question by full formalization, at high cost and with the author’s participation; the ledger answers the form question on the published text and without the author’s cooperation. The mathematical library *mathlib* [27] is the infrastructure in which such formal contracts are stated and reused; the criterion operates one level earlier, on natural-language arguments that have not been formalized. Buzzard–Commelin–Massot [8] document what formalizing modern arithmetic geometry costs in practice; that cost is the

reason a lightweight form criterion has a place beside full formalization rather than in competition with it. Hales et al. [22] verified the Kepler conjecture by machine after conventional review had reached its limit; the criterion is designed to apply *before* review concludes, so that the contested location is nameable at the time the dispute arises.

3.3 Certificate Interfaces in Computer Science

This field contains the closest heuristic analogue among the structures compared here. A certifying algorithm in the sense of McConnell et al. [28] returns, together with its output, a witness that a simple and independently checkable verifier accepts; correctness of the individual run is thereby established without trusting the implementation. The bridge contract has a heuristic interface analogy to certifying algorithms: both place an inspectable artifact beside a claim. The analogy stops before correctness certification. The differences are summarized in Table 4:

Table 4: Structural comparison between certifying algorithms and bridge-contract audits.

Feature	Certifying algorithm	Bridge-contract audit
Artifact producer	the algorithm or producer	a third party, after the fact
Time of issue	with the output	after publication of the text
Checker	formal and simple	subject-matter and hermeneutic
Decidability	typically algorithmic	not guaranteed
Guarantee	correctness of the run or output relative to the checker	textual contract completeness only
Dissent	not a necessary component	explicitly possible and to be reported

Necula’s proof-carrying code [32] presupposes a policy agreed in advance, against which the producer supplies a machine-checkable proof of compliance. The precedent case treated in Section 6 is exactly one in which no such policy was ever agreed: the ledger reconstructs the missing agreement after the fact and names the row at which it was never concluded. Blum–Kannan [7] check results at run time, repeatedly and cheaply, over a stream of instances; the criterion is applied once, to a single argumentative location in a text.

3.4 Assurance Cases and Goal Structuring Notation

Rushby [35] asks when a structured argument carries and supplies the vocabulary for the failure modes that matter here — defeaters, and confirmation bias in the party constructing the argument — assessing persuasiveness in graded, holistic terms. The verdict grammar of Section 2.3 is deliberately coarser and more local: each row receives one of three verdicts about one contract

clause, with the confidence qualifier recorded separately from the verdict itself. The Goal Structuring Notation [4] standardizes how claim, argument and evidence are kept apart, and is domain-neutral by design; the seven contract components of Section 2.1 are the complementary choice, substantively determined for one recurring situation and therefore not transferable without adaptation. Bloomfield–Rushby [6] make the explicit treatment of defeaters obligatory, and do so for a certification context in which an argument authorizes deployment. The criterion carries no admission authority of that kind: it is a reading instrument, and its verdicts are addressed to readers of a proof rather than to a regulator.

3.5 Error Analyses and Controversy Methodology

Aberdein [1] explains why deep disagreements in mathematics persist even among competent parties, working at the level of the disagreement itself. The criterion works one level below: it localizes the individual transported line at issue and takes no position on it. The pair Scholze–Stix [37] and Mochizuki [29] is the reference case of the field and is informative only as a pair — a location-precise error report and its rebuttal, in which the two sides argue past one another for want of a shared grid of form. The ledger of Section 2.3 is the generalized and transferable version of such a grid. Castelvechi [9] documents the institutional consequences of that impasse as they became visible outside the field. Joshi [25] reports on the controversy from the position of a party to it; it is cited here as a party statement and not as independent confirmation of any reading, including our own.

Beyond these, we did not identify an independent peer-reviewed detailed analysis of the abc/IUT controversy in the bounded, non-systematic corpus used for this paper; this is not a systematic-review finding. What that corpus does contain is party primary literature, journalistic reporting [9], and general theory of deep disagreement [1]. We record the absence as a standing gap rather than one the present work closes, and we do not claim to have searched exhaustively. What the criterion contributes is orthogonal to it: a form verdict on a single transport step, not a historical account of the debate (Section 9).

3.6 Position of the Criterion

The criterion sits at the intersection of these five fields rather than in succession to any one of them. It borrows the inspectable-artifact interface from the certifying-algorithms tradition without its correctness guarantee, the explicit separation of claim, argument and evidence from assurance-case practice, the question of the admissible omission from the proof-auditing literature, and the reading of a contested transport step from the documented controversy — and applies them to a class of arguments for which full

formalization is available in principle but, at present cost, rarely in practice. Its verdicts remain of one kind throughout: a served ledger is necessary and not sufficient, and an unserved row localizes a burden rather than refuting a claim.

4 Two Complementary Gate Types

The contract audit of Section 2 is a *logical* gate: it examines a fixed text against a fixed list of obligations, and its verdicts are statements about that text. In the program from which this series grew it is complemented by an *empirical* gate: a cross-route consistency register (a “shadow registry”) that records, for every route under investigation, which statements are actually claimed and where two routes would contradict one another if both were taken at face value, with admission testing against certified views on a graded evidence scale. The instrument side of that register is described in the Zookeeper paper [21]; here it matters as the complementary gate and for its limit, the *consistency trap*: agreement of many diagnostics is evidence of coherence, never of truth — a registry can be consistently wrong. The two gates fail differently, and that is their joint value: the logical gate catches unserved obligations in a single text; the empirical gate catches contradictions *between* texts and runs. Neither replaces proof. The two gates are complementary, not composable: they keep separate outputs, there is no shared decision algorithm that combines them, and no aggregate truth verdict or overall pass is formed from them. A claim examined by both has been examined in two different ways and still owes its mathematics.

Figure 2 contrasts the logical and empirical gate pipelines, illustrating their diagnostic engines, characteristic failure modes, and the non-composability demarcation. Table 5 provides a synoptic taxonomy contrasting the logical gate (Br_ξ), the empirical shadow registry, interactive theorem proving, and certifying algorithms across their scope, trust base, output, and failure modes.

5 A Protocol-Frozen Retrospective Two-Case Illustration (v1-T)

We illustrate the frozen v1-T instrument retrospectively on two historically decided cases: a *retrospective known-gap case*, a proof attempt whose gap was later confirmed by its author, and an *accepted control*, an accepted proof with a structurally comparable transport step. The protocol — row set, five-code handbook, confidence scheme, independent second coding with an explicit dissent rule, hindsight controls, and the success and falsification criteria — was written down and frozen before any source was read; the audits changed none of it.

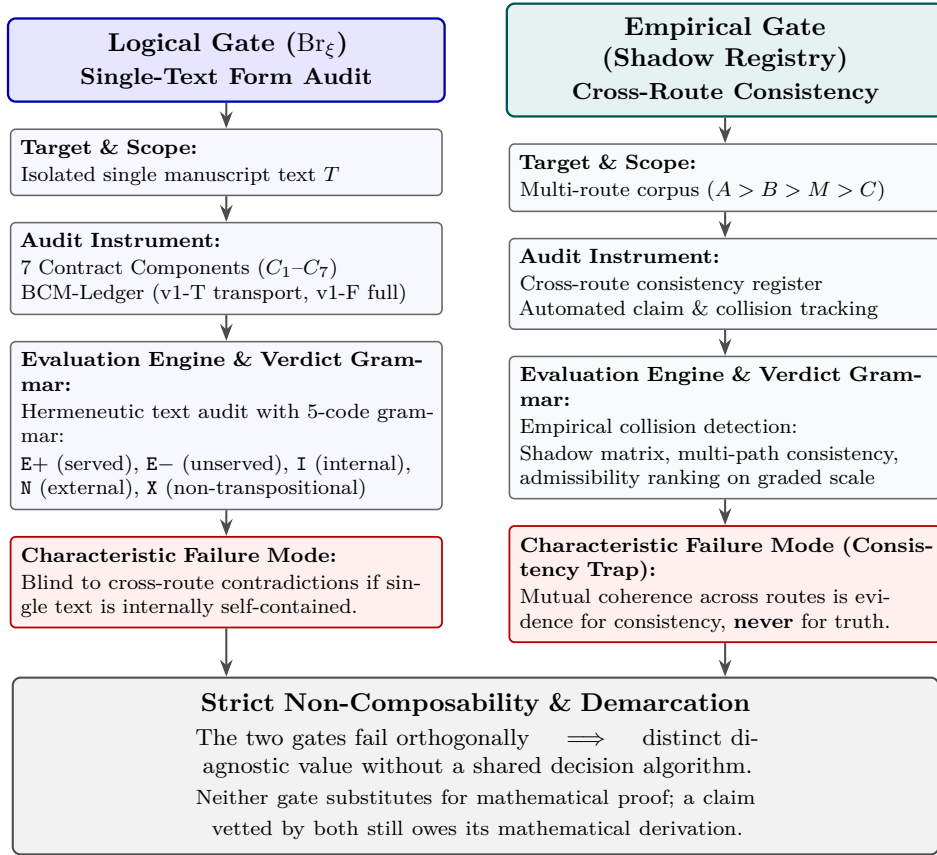


Figure 2: Dual-gate proof assurance architecture: Logical Gate (Br_ξ) vs. Empirical Gate (Shadow Registry).

Table 5: Taxonomy of proof assurance and auditing gates in advanced mathematics.

Gate / Paradigm	Scope of Audit	Verifier / Trust Base	Output Format	Diagnostic Strength / Limit
Logical Gate (Br _ξ / BCM)	Single transport step in mathematical text	Hermeneutic / domain expert	5-code contract ledger (E+, E-, I, N, X)	Localizes exact unserved clause; judges form, not truth.
Empirical Gate (Shadow Registry)	Cross-route claim consistency across corpus	Automated registry lookup ($A > B > M > C$)	Consistency / collision matrix	Detects cross-text contradictions; subject to consistency trap.
Machine Formalization (Lean / Coq)	Deductive kernel across complete proof	Mechanized proof assistant	Binary validity certificate	Absolute deductive certainty; high formalization cost.
Certifying Algorithms (PCC / Runtime)	Algorithmic input-output pair	Algorithmic witness verifier	Computational certificate	Decidable instance checking; requires pre-agreed policy.

That temporal priority is documented internally, not externally. The protocol was not publicly registered before coding; its temporal priority is documented by the dated internal project record and the audit notes. We therefore use “protocol-frozen”, not “externally preregistered”. Where this paper says a choice was fixed in advance, it means: prospectively frozen in the project record before the two source-coding passes. A checksum computed today would establish the identity of the file published today, not the date on which it was frozen.

5.1 Design and Instrument Version

The two historical cases were coded with **BCM-Ledger v1-T**, the six-row transport instrument frozen on 15 August 2026 before source coding. Its version-scoped rows are T-R2a, T-R2b, T-R4a, T-R4b, T-R6, and T-R8, with the case-neutral semantics reproduced in the table below. This is not the full v1-F ledger of Section 2.3; the two versions are related only by the published crosswalk (Section 2.4).

Table 6: The frozen six-row transport instrument (BCM-Ledger v1-T) and case-neutral test questions.

v1-T row	Case-neutral question of the frozen instrument
T-R2a	Is the transport or transfer law written out explicitly, as a formula or statement with an address?
T-R2b	Are the transported quantities carried with a visible conversion factor?
T-R4a	Is the compensation or normalization mechanism named and located?
T-R4b	Is its existence and well-definedness proven in the text, not merely asserted?
T-R6	Are second modes or case distinctions displayed and treated symmetrically?
T-R8	Is the chain-closing compatibility derived rather than posited, and at which level (local or globally summed)?

The audited unit in each case is the central transport step, instantiated over these six rows. Each row receives exactly one of the five codes of Section 2.3. Each case was coded twice: a first audit and an independent second coding whose auditor received only the frozen protocol and the source, not the first coding. Success was fixed in advance: in the known-gap case, at least one row must fire ($E-/N$) and its address must match the historically documented gap; in the control, no row may fire (an I is non-critical). Failure modes were fixed symmetrically: no firing, firing at the wrong address, or a firing in the control.

5.2 The Retrospective Known-Gap Case: The 1993 Fermat Announcement

The first case is Wiles’ 1993 proof announcement, audited through the contemporaneous technical report of Rubin and Silverberg [34]; the audit therefore concerns the 1993 state of the argument *as reported*, a scope limitation carried explicitly. Both coders, independently, produced the same picture: the transport law and its conversion quantities are explicit (T-R2a, T-R2b = $E+$), the compensating mechanism — the Euler system — is named and located (T-R4a = $E+$), the case distinctions are displayed (T-R6 = $E+$), and exactly one row fires concordantly: *the existence of the named mechanism* (T-R4b = $E-$), at the passage where the report states that the construction of a suitable Euler system was “not yet complete.” The chain-closing row T-R8 was coded $E-$ by the first coder (as a consequence of T-R4b) and I by the second; under the double-coding rule below the dissent is verdict-neutral, since the success criterion was already met by T-R4b alone.

The address of the firing row is the passage that the report itself marks as incomplete, and it is the obligation that the subsequent repair of the argument had to discharge; the surrounding chain was not at issue there. The diagnostic content lies as much in the four rows that did *not* fire: the located gap is narrow, not diffuse.

5.3 The Accepted Control: An Accepted Transport Step

The second case is Prasad’s volume formula [33], whose local-to-global normalization step is an accepted instance of the same transport shape. Both coders found no firing row: five rows explicitly discharged in both codings; one row (T-R6) split between I (first coder, on account of the author-declared function-field asymmetry: the relevant convergence statement is proved over number fields by an external reference, while for global function fields the text records only that “a similar proof applies” ([33], §3.2)) and $E+$ (second coder), which is verdict-neutral because I is non-critical by the frozen protocol. The scope reservation admitted by the protocol — an audit of a transport step inside an accepted proof rather than of the whole paper — did not have to be invoked as an X . The control also supplies a positive image of what the rows measure: the normalization constant is carried as a visible factor rather than silently set to one, and the chain closes through an addressed chain of equations rather than an assertion that it “follows formally.” This is the only accepted control in this paper.

5.4 The Complete Code Matrix

Both codings of both cases are reported in full in Table 7. Columns F1 and F2 are the first audit and the independent second coding of the known-gap

case (Wiles 1993, as reported); columns P1 and P2 are the first audit and the independent second coding of the accepted control (Prasad). No cell below is a consensus value.

Table 7: Complete double-coded audit matrix for the retrospective test cases (Wiles 1993 vs. Prasad 1989).

v1-T row	F1 (Wiles 1st)	F2 (Wiles 2nd)	P1 (Prasad 1st)	P2 (Prasad 2nd)
T-R2a	<i>E+</i>	<i>E+</i>	<i>E+</i>	<i>E+</i>
T-R2b	<i>E+</i>	<i>E+</i>	<i>E+</i>	<i>E+</i>
T-R4a	<i>E+</i>	<i>E+</i>	<i>E+</i>	<i>E+</i>
T-R4b	<i>E−</i>	<i>E−</i>	<i>E+</i>	<i>E+</i>
T-R6	<i>E+</i>	<i>E+</i>	<i>I</i>	<i>E+</i>
T-R8	<i>E−</i>	<i>I</i>	<i>E+</i>	<i>E+</i>

Inter-coder concordance was 10 of 12 case-row comparisons. The two dissents are F/T-R8 and P/T-R6; neither was harmonized. The rule under which these codings were read is the following.

A critical firing in the known-gap case counts toward the pre-specified success criterion only when both coders assign *E−/N* to the same row. A control failure is recorded if either coder assigns *E−/N*. Discordant code pairs remain visible and are never replaced by a synthetic consensus code.

Under this rule exactly one row counts as a concordant critical firing in the known-gap case, namely T-R4b; in the control no coding contains *E−* or *N* at all. Stating the rule explicitly is a disclosure of the rule already applied, not a retrospective change to the codings.

5.5 Anchoring and Availability

Every coding in both audits carries a verbatim quotation with a page or section anchor from the named edition, together with the three-axis confidence record prescribed by the frozen protocol: source binding, criterion fit, and substance status, each on a three-point scale. Those per-row anchor tables live in the dated project record — the frozen protocol, four audit notes (a first audit and an independent second coding for each case), and the operator synthesis that carries the verdict wording. This draft reproduces the complete code matrix but not the per-row anchors.

5.6 What the Illustration Shows—And What It Does Not

Under the pre-specified criteria, the two pre-specified case criteria were met in this illustration: the confirmed gap of the 1993 argument is located

at a single concordantly firing row whose address matches the historically documented gap, and neither coding assigned $E-$ or N in this one control case. We state the outcome in the formulation fixed in the protocol: the two cases are *consistent with the certificate's design intent*. They are one retrospective known-gap case and one accepted control — an illustration, not a validation statistic, and not a measurement of any rate. The exercise is a retrodiction, not a prediction: the historical outcome was known to both coders, and its value rests entirely on the row assignment having been fixed in writing before reading, with the falsifiable alternatives (wrong address, diffuse firing, a firing in the control) stated in advance and not realized.

Two further limits belong here. The illustration concerns v1-T only; it does not exercise the full v1-F ledger of Section 2.3, and nothing below inherits support from it. And with two cases and two coders, the reported concordance describes this pair of audits; it is not an estimate of the instrument's reliability.

6 The abc Instantiation

6.1 The Gate Case

This section reports the one instantiation contained in this paper. It is a *selected-rows* instantiation carried out with rows of **BCM-Ledger v1-F** (Section 2.3): the rows exercised below are F-R2a, F-R2b, F-R4a, F-R4b, F-R6a, F-R6b and, where its trigger fires, F-R8; the remaining rows (F-R0, F-R1, F-R3, F-R5, F-R7) are not coded here, and a coding of the complete matrix for these cases is deferred to a later version. The row labels below carry the prefix F-, and none of them is the row of the same number in the frozen transport ledger v1-T used in Section 5.

The step at issue is a substep in the proof of Corollary 3.12 in the third volume of inter-universal Teichmüller theory [30]. Its conclusion has the form axiomatized in Definition 2.1: a log-volumetric reading $\nu_q(x)$ of the q -pilot object $x \in X_q$ is asserted to lie in a target region $H_\Theta = \mathbb{R}_{\leq -|\log \Theta|}$ fixed on the Θ -side, where the two sides share no common evaluation by construction. The step labeled (xi-e)→(xi-f) is where that membership is drawn.

That this step carries the load is the result of Part 1 [11] and is taken as given. The criterion asks a different question of the same passage: not where the argument is thin, but which contract components the presentation serves. Everything below is read in the register of Section 2.5 — a row that does not deliver names a clause a complete argument must supply, and says nothing about whether it can be supplied.

6.2 The Audited Substep of the Original Argument

The audit covers the rows activated at the substep itself: F-R2a, F-R2b, F-R4a, F-R4b, F-R6a, F-R6b and, by the multi-mode trigger, F-R8. The

remaining rows of the ledger carry no verdict here. The table below collects the verdicts; what follows are the grounds.

At the audited substep no intra-link transport operation is performed; F-R2a is therefore X at that substep, not E+. The cross-framework numerical identification needed by the conclusion is assessed separately under F-R2b and F-R6a. A common evaluation in C does not discharge Component 6 unless the required compatibility identities are present and addressed. What supports the first of these statements is that the parallel-transport mechanism is described as one that “does not consist of a simple instance of transport of some set-theoretic region”, and that the q -side readout is declared “coric”/“fixed” and expressly not subject to the indeterminacies the Θ -side carries: the readout keeps its identity by stipulation and by the absence of a transport step, not by a proved relation.

F-R2b partially delivers, and no more than that. At (xi-d) the two sides are brought into one evaluation, only thereby yielding what the text calls “completely comparable objects”, and global arithmetic degrees are read as log-volumes on both sides. That is an identification carried by a shared reading convention rather than by a visible conversion factor or a compatibility equation, and the crosswalk caps a row of that shape at I. The row reaches that ceiling rather than falling below it because the identification is itself exhibited at a named substep and can be read off the text: what the documented reconstruction lacks is a derivation of the identification, not the identification. At this step the row is moreover not separable from F-R6a, the identification of the two evaluation containers being part of the bridge step itself.

F-R6a does not deliver. A numerical evaluation ν_C on the container is exhibited; what the hardened Component 6 requires in addition — the compatibility identities the claim needs, in particular $\nu_C \circ i_q = \nu_q$ on the stated source domain — is not derived at the audited step. The source-side reading is held fixed by stipulation — the passage quoted under F-R2a above, which declares the readout “coric”/“fixed” and exempt from the Θ -side indeterminacies — and a stipulated invariance is not the identity Component 6 asks for. The row would deliver partially only if a documented reconstruction of that identity were supplied; this audit supplies none. The earlier reading of this passage, which recorded Component 6 as served, rested on the weaker either/or formulation that Definition 2.1 no longer contains.

F-R6b does not deliver, and it is the principal chain-closing obligation. The membership is not derived from the admissible linkage but obtained from a condition imposed on the admissible operations: the inclusion “then follows formally” from the requirement that the construction of the output possibilities constitutes, in particular, a construction of the input readout, and a summary of the same argument speaks of “two tautologically equivalent ways” to compute one log-volume. Where the named order lemma of Component 7 would stand — with its hypotheses, a claim of applicability

at this step, and a proof address — a pair of declared properties of the algorithm’s output stands instead. The row records the absence of such an addressed lemma. It does not record a judgment on whether one could be proved.

Two fairness statements belong to that verdict rather than beside it. Confidence is high precisely because the load-bearing passages use those words themselves; and “tautological” is used affirmatively there, the text elsewhere calling its conclusions nontrivial albeit essentially tautological. For that author the tautologicity is the point, not an admission, and reading it as self-refutation would be an authority argument in reverse.

The load rows require the second split. In the audited volume the load carried by the theta values q^{j^2} is named but not quantified — what is booked at the identified site are labels rather than magnitude — and at the bridge step the internal structure is deliberately set aside. The quantification is not omitted but declared, its address given as the explicit log-volume computation in §1 of the fourth volume [31], and that address pays out: the load is carried through that computation to the closed form $(\ell+1)/24$ and stands in the final bound as $-\frac{1}{6}(1 - 12/\ell^2) \log q$. F-R4a therefore delivers for the composite of the two volumes; F-R4b does not, that section taking the corollary’s conclusion as an input and quantifying downstream of the bridge. Both rows are reported at Component 5, which is where the crosswalk localizes the load budget. An early concern recorded in the ledger note of May 2026, that the j^2 factors might vanish in one coordinate and return unbooked, does not hold for the composite and is withdrawn: booked elsewhere as announced is not the same finding as unbooked.

F-R8 is triggered here, since the claim’s domain model contains an alternative mode that bears on the order relation. It partially delivers: the second mode is named, the vertical-shift conflict stated and its price declared, but it is converted beforehand into a one-sided inequality and suspended at the step itself.

The audited rows therefore do not reduce to a single surviving obligation. F-R2a is out of scope at the substep, F-R2b partially delivers, and F-R4a delivers for the composite of the two volumes. Against that profile the remaining obligations are weighted as follows. The audit distinguishes one principal chain-closing obligation from three additional findings: F-R6b is the principal order-lemma obligation; F-R6a records the missing compatibility identity of the common evaluation; F-R4b records a separate booking-at-the-bridge deficit; and triggered F-R8 records partial treatment of the second mode. The ledger therefore localizes differently weighted findings rather than a single surviving row.

6.3 A Comparative Discrimination Case: Joshi’s ATS Program

A criterion that returns the same verdict everywhere is a template, not an instrument. We therefore apply the same selected v1-F rows, with unchanged row semantics, to a second program of the same dispute complex reaching the same inequality by other means: Joshi’s arithmetic Teichmüller spaces [23, 24]. This case is not a positive control and supplies no independent confirmation of the inequality. It is a methodologically distinct, preliminary comparison used to ask whether the same selected rows produce a different pattern of served, reconstructed, and unserved obligations. The only accepted control in this paper is the one in Section 5. It does: the profile differs, and the remaining obligation has a different address and a different shape.

F-R2a rests on a proved relation rather than a stipulation: the scaling relation between the valuations along the program’s own link is a theorem there, and the j^2 factor is carried visibly into an explicitly defined log-volumetric scale. Here the split separates cleanly, which is why this audit produced it. F-R2b, by contrast, rests on the choice of the same global normalization on both sides — a convention, not an exhibited conversion factor — and therefore partially delivers at most, exactly as the crosswalk caps a row of that shape. It reaches that ceiling because the normalization is explicitly chosen and stated, so the reconstruction is documented; what is absent is a conversion factor carried visibly through the step.

The load obligation was audited before the split was introduced and is therefore reported as a single verdict, at F-R4a. The load is booked against a global constraint, the product formula, with high confidence for the address and medium for completeness, the compensating renormalization being inferred rather than constructed. The address is served and completeness is not established, so the row delivers for the address and only partially for completeness — not unqualifiedly. An audit at the split would have to separate the served address from the inferred existence; this one did not, and F-R4b therefore carries no separate finding in this case.

F-R8 is delivered as the row asks: a global Frobenius shift is named, deployed asymmetrically for the two bounds, and flagged by the author as a correction to his own earlier version — and because it is visible, the remaining obligation has an address at all.

F-R6a partially delivers, and the shape of the remainder is this case’s yield. The order relation carrying the conclusion holds between sums over all places; place-wise it is expressly disclaimed, the author stating that local term-by-term comparisons at each prime cannot be made without violating the concurrent normalizations, and the theorem’s own remark holding that this choice “precludes direct comparison of various local quantities”. The identification of the two evaluations rests on identical global normalization and is called a tautology at that point. That word must be scoped: it occurs

at several places in the program with different reach, only its use for the middle equality of the carrying theorem is load-bearing here, and it is not the author calling his own arithmetic proof of the corollary a tautology.

F-R6b is where this case differs most sharply in form from the original argument, and where the auditor’s boundary must be observed most carefully. What the row tests is present: a named theorem with stated hypotheses, an assertion of its applicability at the step, and a written-out proof with an address. That is the whole of what a served F-R6b certifies. Against it stands a substance finding, reported here at F-R6b because that is where the obligation it concerns is localized: the substance audit of Part 5 [14] typed the core step of that proof as open against the construction it cites, so the core of the obligation remains open. That finding is not a form verdict, and the auditor does not adjudicate it.

Two corrections revised the historical reading of this case in opposite directions. Within the particular normalization framework named in the audited Part 4 [13], the identification was read as definitional rather than a concealed choice; this is not an unconditional identification across ATS fields, place ranges or volume constructions. Set stability and the pinning of the measure remain separate obligations. The Part 5 substance finding [14] is distinct from that form reading. The rigidity result cited in the historical discussion [15] is conditional on its stated field, all-place and normalization hypotheses; it removes no such hypotheses and supplies no unconditional ATS or volume conclusion. Table 8 retains the historical comparative audit results, scoped to the versions audited. The volume carrying the abc conclusion describes itself as a preliminary version for comments.

The audited unit was these rows at the identified step in both cases; the remaining rows of Section 2.3 carry no verdict here, and a row coded X is not counted against either presentation. What the comparison localizes is a difference in the *form* of the remaining obligation: a pair of declared properties distributed over a remark and several substeps, against a single named equality with a written-out proof. Whether that proof is correct is not a question this instrument answers.

6.4 Inheritance: What Travels with a Citation

A downstream inheritance example moves the question past the audited step. Two application papers by Zhou [38, 39] derive Diophantine consequences resting on the inequality of Corollary 3.12, and the audit asked one question: how the bridge is taken over. In the first it appears as a proposition whose entire proof is one sentence — “The proposition follows from the μ_6 -version of [37], Corollary 3.12”, the reference being to the third volume in that paper’s numbering — and is used exactly once, after which the downstream chain proceeds. In the second, in the version audited, the corollary is not named at all: the bridge is inherited through the first paper and extended

Table 8: Comparative audit profile for the abc transport step (original IUT presentation vs. ATS program).

v1-F row	Original argument	Comparative case (ATS program)
F-R2a	out of scope (X): no intra-link transport operation at the substep	delivers (high): a proved scaling relation along the program’s own link
F-R2b	partially delivers: identification by a shared reading convention; not separable from F-R6a here	partially delivers: the same global normalization, set by convention
F-R4a	delivers (high) for the composite of the two volumes	audited unsplit, reported here: delivers for the address, partial for completeness — the compensating renormalization is inferred rather than constructed
F-R4b	does not deliver (high): quantified downstream of the bridge	no separate finding: not audited at the split (see F-R4a)
F-R6a	does not deliver: no compatibility identity derived at the step	partially delivers (high): order after summation, place-wise disclaimed
F-R6b	does not deliver (high): membership “follows formally”	textually served: named theorem, hypotheses, proof address; core step typed as open by a substance audit
F-R8	partially delivers (high): named and priced, suspended at the step	delivers (high): named, asymmetric, declared

to a new class of initial data by tracking a single hypothesis through the cited volumes, without the bridge step being touched. What both papers contribute of their own lies on the far side of the bridge.

None of this is irregular: citing an established result instead of re-deriving it is what citation is for. What makes the structural point visible is that both papers state plainly where they omit a step of their own. Those reservations attach to their own omissions and none attaches to the imported step — not because the authors declined to make one, but because nothing in a citation carries a contract state. A served row and an unserved row travel identically through a bibliography.

This is the practical case for a contract certificate as a *citation interface*. An audited ledger is a citable artifact: it states, row by row, what the cited result was found to serve and what it was found to leave open, so that an importing paper inherits a known contract state instead of an unknown one. Under such a discipline — formulated in Part 3 [12] as an import ledger — neither paper’s mathematics would change; each would carry one further line naming the row its import leaves unserved. Both findings are scoped to the versions audited, the second paper describing itself as an older version to be updated, and one question the audit left open — whether a normalizing factor in the first paper’s statement renormalizes the same quantity the corollary carries — is left open here.

6.5 An External Point of Comparison

The last item in this section is not ours. An interim report of a formalization project, published in July 2026 [26], isolates a compatibility problem at the same passage — and isolates it in the shape of a contract. Its condition, numbered (9-1) there, requires the existence of a suitable choice of integral structure under which two evaluation isomorphisms with a common target agree: one arising from the q -pilot in its native structure, the other obtained anabelian and Kummer-theoretically and subject to indeterminacies. The report states that the project “has not yet reconstructed a proof of the required compatibility”, and that the proof of the final numerical inequality “hinges on the compatibility (9-1) which is not manifestly false. However, we, the LANA project, do not have a proof of (9-1) at this time.”

Read against Definition 2.1, the proposed correspondence is object-wise, not a verified transfer: a common target quotient, typed maps, readouts, integral structures and indeterminacies would have to be identified with their actual objects, domains, quantifiers and implication direction. The existence of a quotient or a shared target alone does not serve the compatibility identities of Component 6. If an equality of readouts has been established in the common value space V , reflexivity gives the numerical comparison in V , not conversely. This does not by itself give the container order (H2): ν_C need not reflect order. Equality of the identified images in C gives (H2)

by reflexivity in C ; numerical comparison alone requires order reflection or a separately addressed proof of (H2). Membership in H_Θ additionally requires the applicable (H1)–(H5), including the target bound (H5). No such IUT-to-C7 bridge is proved by this comparison.

The ledger note that fixed the criterion in May 2026 recorded a conditional expectation: should a formalization effort carry this passage, the contract would have to appear in it as a lemma. The report has that shape. In this single interim report, at this single passage, an independently formulated compatibility condition occupies a position analogous to F-R6b. We record a *post hoc* structural resemblance, not convergence on the general criterion and not confirmation of our case verdict. No claim of precedence is made either; the analysis is that project’s own and in its own terms. The limits are equally plain: the document is an interim report without a final verdict and records internal dissent, and our mapping is object-wise, not a proof that (9-1) and component 7 are equivalent. The report’s own position — required, unproven, not manifestly false — is precisely what the verdict grammar of Section 2.3 exists to express: a row that does not deliver localizes a burden and refutes nothing.

7 Outlook: RH Uniformity Claims

The criterion is not specific to the abc literature. Positivity approaches to the Riemann Hypothesis contain the same two shapes — transport steps (finite-model results carried to a limit object) and global-only claims (positivity available only after summation) — and the corresponding ledger rows could be instantiated there: uniformity obligations for finite-section bounds, quotient obligations where majorization arguments are excluded, and an explicit payment obligation for archimedean mass. We defer any such instantiation to a dedicated treatment, synchronized with the self-application part of this series, because our own program is among the audited objects there and the protocol-freezing discipline of Section 5 requires the operationalized rows, primary sources, and claim boundaries to be fixed before the section is written. This section is an outlook and not an instantiation: the present paper deliberately contains no RH-side verdicts.

8 A Related Structure: Global-Only Claims

A recurring shape in both instantiation domains is the *global-only* claim: a statement asserted to hold only after summation over all places, with local, place-wise access explicitly unavailable. In Weil-positivity approaches to RH the positivity of a quadratic form is a global sum whose individual terms carry no sign; in the abc literature, the source of the principal repair program restricts its comparisons to the summed level in so many words

(“one cannot make local term by term comparisons at each prime” [23, 24]), and its normalization mechanism is a product-formula constraint — again a statement about a product over all places, not about any single factor. The audit trail for both readings is in Parts 4 and 5 of this series [13, 14].

For the criterion this shape triggers a specific obligation. Where a claim lives only at the summed level, the contract components 6–7 cannot be served place-wise; what must then be supplied is a *contract lemma*. We state what such a lemma has to contain.

Definition 8.1 (Global-only contract lemma). *Let $(x_v)_{v \in \Sigma}$ be an admissible family, let $G((x_v)_v) = e$ be a stated global constraint, and let $S((x_v)_v)$ be the aggregate used by the claim. A global-only contract lemma is a named, addressed statement that specifies (G1) the admissible class and convergence/domain conditions, (G2) the exact global constraint, (G3) the aggregate equality or order implication needed for the claim, and (G4) the local degrees of freedom that remain after imposing the constraint. It discharges F-R6b only at the global level. A place-wise rigidity conclusion requires an additional named injectivity, strictness, or uniqueness hypothesis and may not be inferred from globality alone.*

In this paper the definition is an outlook interface, not a second instantiation; no global-only ledger verdict is reported. The tool supplement of this series [19] carries out a construction of this kind for the global normalization of the abc repair program, and the rigidity theorem of Part 6 [15] exhibits one case in which a global multiplicative constraint over all places, together with the hypotheses stated there, does force place-wise rigidity — which is an instance of (G4) discharged under named additional hypotheses, not a general license to read rigidity off globality. Within v1-F the triggered row F-R8 is activated at the global level only when several global modes or branches can affect the order of the aggregate. The lesson for auditing is symmetrical: global-only is not a defect, but it converts local obligations into a single named global one — and the audit must check that this one is discharged, not dissolved into notation.

8.1 Current Developments in Parts 4–6

The later v0.3 papers are additional, version-specific research references, not replacements for the v0.2 pins used in the retrospective audit above. Part 4 [16] retains a partially documented normalization convention while separating the typed log-image, hull and log-volume contracts. Part 5 [17] distinguishes raw and normalized coordinates and requires the actual common fixed coordinates, coefficient transformation and period-image action for its Frobenius route. It also records the absolute-value bars actually printed in the narrowly checked ATS-III corollary; the earlier atomic-sign exculpation is withdrawn under the explicit Tate-place and finiteness assumptions there.

These later findings do not recode the historical rows or decide the whole ATS, abc or IUT chain.

In the standard-modulus normalization of Part 6 [18], local degree factors are included so that

$$\sum_{v \in M_L} \log b_v(x) = 0 \quad (x \in L^\times).$$

For one fixed number field L , all its places including the archimedean ones, real weights independent of x , and every $x \in L^\times$, writing the local norm realization as $\rho_v = b_v^{a_v}$, the weighted product formula concerns the final effective weights

$$\beta_v = a_v \lambda_v, \quad \sum_{v \in M_L} \beta_v \log b_v(x) = 0 \quad \text{for all } x \in L^\times.$$

Under these hypotheses rigidity gives $\beta_v = c$; it does not identify the raw λ_v with c when degree factors are separate. A constant global factor, including j^2 , is allowed. Tests restricted to $L^\times$ at places of an extension L' do not determine the individual L' weights. Common field, norm and freeze identifications remain prerequisites for a cross-branch comparison.

For a quantity consumer, the required family is linear:

$$Q_y = \sum_v \beta_{y,v} q_v, \quad Q_{\text{ref}} = \sum_v q_v, \quad Q_y = c_y Q_{\text{ref}},$$

with y -independent q_v and finite sums or justified absolute convergence. The last equality is conditional on the same effective-weight hypotheses; $Q_{\text{ref}} = Q_{\text{cl}}$ is an additional classical-reference bridge. Pointwise normalization (5.3.3) yields the classical degree on $L^\times$ and height on $\mathbb{P}^n(L)$ under its stated identification, without an automatic algebraic-extension or line-bundle-height transfer. Membership and nonmembership of measures and log-volumes in this class remain unknown; an arbitrary ATS hull is not silently identified with a maximal-order module hull. These are current scope qualifications, not new F-R6 verdict codes.

9 Limitations and Fairness

Four limitations bound everything above. *First*, the criterion judges form. No verdict in this paper or in the series is a truth verdict: not about abc, not about IUT, not about any audited program. An unserved ledger row is an obligation with an address; theorems are refuted by mathematics, not by audits. What the auditor certifies is narrower still: that an obligation is present, typed and addressable in a named version, never that the mathematics it names is sound. *Second*, the audit frame is neutral in the underlying controversy. The same rows are applied to every audited text — the original

program, its critics’ reading, and the repair program — and the retrospective illustration of Section 5 was designed with a symmetric falsification criterion precisely so that the instrument could have failed. *Third*, the evidential basis is small by construction and version-scoped: one retrospective known-gap case and one accepted control under v1-T, one abc instantiation with one comparative discrimination case under v1-F, plus a downstream inheritance example and an external comparison. Nothing in the v1-F instantiation inherits support from the v1-T illustration. We use the formulation fixed in the protocol, “consistent with design intent”, and avoid “validated” throughout. *Fourth*, a fairness protocol governs the practice: a verdict that depends on the audited author’s own wording carries that wording as a verbatim quotation with an address from a named version; every other verdict argued in the running text is carried by a paraphrase of the passage or feature relied on in the named version, or by a citation into the published series part that documents it; the row-code matrices are summaries of archived audit records whose per-row anchor tables are not yet published — until the planned supplement appears, the matrices are to be read as summaries of those records, not as independently checkable claims (Section 5.5); confidence is stated in a fixed three-axis scheme; findings of the form “the text does not show” are never promoted to “this is false”; and audited authors’ own scope statements are quoted, not paraphrased, wherever a verdict depends on them.

The two retrospective cases also use different source genres and audited versions: the Wiles case is accessed through the contemporaneous report by Rubin–Silverberg, whereas the Prasad case uses the primary paper. Their raw v1-T pairs remain visible: the Wiles T-R8 pair is $(E-, I)$ and the Prasad T-R6 pair $(I, E+)$. This source asymmetry and the 10/12 agreement do not estimate reliability, discrimination or a false-alarm rate; neither pair is silently converted into a v1-F F-R8 code. The LANA comparison still requires its separate object, domain, quantifier and implication-direction bridge; no equivalence or confirmation follows from structural resemblance.

10 Conclusion

The current illustration now preserves the full conditional C7 contract; the incomplete earlier diagram has been corrected without changing the definition or the raw retrospective codes. Current Parts 4–6 refine the remaining normalization and quantity-consumer obligations without supplying a general methodological validation or closing the field, period, volume or LANA transfer bridges.

We have stated a checkable form criterion for transport arguments: seven contract components, a versioned ledger with a three-verdict grammar over five raw codes, and an audit practice with verbatim anchoring and evaluation

criteria fixed before coding. Its position among existing structures is specific — the closest heuristic analogue is the certifying-algorithms tradition, but the artifact here is reconstructed by a third-party auditor rather than issued by the prover, and what a check establishes is textual contract completeness rather than correctness — and, within the bounded corpus surveyed here, we found no work occupying the same place in the mathematics-controversy literature.

The contribution established here is one abc instantiation of a form-audit criterion. The two historical cases illustrate v1-T retrospectively; they do not validate the full v1-F instrument. RH and global-only applications remain future work. In the protocol-frozen two-case illustration, the pre-specified criteria were met. Applied to the selected abc transport step, the ledger replaces a diffuse disagreement about that step with a profile of named obligations, weighted and each with a citation; it does not adjudicate the controversy that surrounds it. What the criterion cannot do is prove anything, and a served row certifies no mathematics. Its ambition is smaller and, we believe, useful: that disagreements about transport claims can be conducted as disagreements about named contract clauses — checkable, quotable, and revisable when the text changes.

AI Disclosure

AI systems were used extensively in all phases of this work: conception and research discussion, literature and source work, mathematical analysis and proof development, creation and checking of the computational scripts, execution and evaluation of the computational runs, text, translation, and typesetting.

Quality assurance: use of several independent models from different providers with counter-checks and adversarial AI reviews; traceable, versioned computational scripts with stored result files and checksums; continuously maintained proof and register documents; strict claim hygiene (assertions only with documented evidence, addenda instead of silent corrections); changelogs across all versions.

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